

Saturation of Resolution Geometry

A Missing Invariant: Disproof and Repair

Matěj Rada

MatejRada@email.cz

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Abstract

Having drawn resolution geometry as a stratified orbit-space bundle, we can test whether its stated coordinates capture all of its structure. The test is saturation: is the fibre coordinate list—resolved spectrum λ , null spectrum μ , and canonical correlations ρ —a *complete* invariant of a resolution geometry up to gauge? We prove it is not. A dimension audit shows the gauge quotient of the fibre has dimension $p + q + pq$, while (λ, μ, ρ) supplies only $p + q + \min(p, q)$, leaving a gap of $pq - \min(p, q)$ that is strictly positive once $\max(p, q) \geq 2$. We exhibit an explicit counterexample: two resolution geometries with identical (λ, μ, ρ) but different gauge invariant $\text{tr}(\Sigma_R C \Sigma_N C^\top)$, hence not gauge-equivalent. The missing structure is identified: the *frame alignment* between the eigenframes of the covariance blocks and the singular frame of the coupling—geometrically, ρ records the magnitude of the coupling (the principal angles), and the missing invariants record its *orientation* relative to the uncertainty axes. We then repair the coordinate list: the enlarged set $\{\text{tr} \Sigma_R^k, \text{tr} \Sigma_N^k, \text{tr}(\Sigma_R^a C \Sigma_N^b C^\top)\}$ has exactly $p + q + pq$ members, and its differential has rank equal to the quotient dimension at a generic point, so it locally separates orbits—the fibre *re-saturates*, with no further hidden continuous invariants. Finally we reconcile with the existing corpus: on the isotropic (maximally degenerate) stratum ρ is complete, which is the precise sense in which “canonical correlations classify the coupling on a stratum” was true; the gap is a top-stratum phenomenon. The saturation test therefore did what such tests are for: it found a genuine missing core invariant and fixed it. Every claim is numerically verified.

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1 The saturation test

A geometry is only as trustworthy as the completeness of its invariants. Once the object is in hand—here, the stratified fibre of a resolution geometry—one can ask the sharp question: do the coordinates we wrote down determine the object, or is there structure they miss? If they are complete, then *saturating* the geometry (varying it freely within its coordinates) exposes no unaccounted-for degrees of freedom; if they are incomplete, the shortfall appears as gauge-invariant quantities that the coordinates cannot reconstruct. This paper runs that test on the fibre coordinate list and reports the outcome, which is negative and then repaired.

Definition 1.1 (The fibre and its claimed coordinates). Fix resolved dimension p and null dimension q . A resolution geometry (at a point) is a triple (Σ_R, C, Σ_N) with $\Sigma_R \in \text{Sym}^+(p)$, $\Sigma_N \in \text{Sym}^+(q)$, $C \in \mathbb{R}^{p \times q}$, taken modulo the gauge group $O(p) \times O(q)$ acting by

$$(\Sigma_R, C, \Sigma_N) \mapsto (U\Sigma_R U^\top, UCV^\top, V\Sigma_N V^\top), \quad (U, V) \in O(p) \times O(q).$$

The coordinates proposed for the fibre are the resolved spectrum $\lambda = \text{eig}(\Sigma_R)$, the null spectrum $\mu = \text{eig}(\Sigma_N)$, and the canonical correlations $\rho = \text{singular values of } \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$.

The question is whether (λ, μ, ρ) determines the gauge class.

2 Disproof I: the dimension audit

Theorem 2.1 (The claimed coordinates are too few). *The generic gauge quotient of the fibre has dimension $p + q + pq$. The coordinate list (λ, μ, ρ) supplies $p + q + \min(p, q)$ numbers. Hence for $\max(p, q) \geq 2$ there is a positive saturation gap*

$$\Delta = pq - \min(p, q) = \min(p, q) (\max(p, q) - 1) > 0,$$

and (λ, μ, ρ) cannot be a complete invariant.

Proof. The parameter count is $\dim \text{Sym}(p) + pq + \dim \text{Sym}(q) = \frac{p(p+1)}{2} + pq + \frac{q(q+1)}{2}$. On the top stratum (Σ_R, Σ_N with distinct eigenvalues) the gauge action is generically free, so the generic orbit has dimension $\dim(O(p) \times O(q)) = \frac{p(p-1)}{2} + \frac{q(q-1)}{2}$. Subtracting,

$$\dim(\text{quotient}) = \left(\frac{p(p+1)}{2} + pq + \frac{q(q+1)}{2} \right) - \left(\frac{p(p-1)}{2} + \frac{q(q-1)}{2} \right) = p + q + pq.$$

The coordinate list has p (for λ) + q (for μ) + $\min(p, q)$ (for ρ) entries. The difference is $pq - \min(p, q) = \Delta$, positive whenever $\max(p, q) \geq 2$. $\square$

Verification 2.2 (Quotient dimension, measured). Computing the rank of the infinitesimal gauge action $(A, B) \mapsto ([A, \Sigma_R], AC - CB, [B, \Sigma_N])$ at a generic point and subtracting from the parameter count gives the quotient dimension; it matches $p + q + pq$ and the gap Δ exactly:

(p, q)	quotient dim = $p + q + pq$	(λ, μ, ρ) count	gap Δ
(1, 1)	3	3	0
(2, 2)	8	6	2
(3, 3)	15	9	6
(2, 3)	11	7	4
(4, 4)	24	12	12

Only the trivial case $p = q = 1$ is saturated by (λ, μ, ρ) . Everywhere else the coordinates are strictly too few.

3 Disproof II: an explicit counterexample

The dimension audit forces incompleteness; we make it concrete by exhibiting two inequivalent geometries the coordinates cannot tell apart.

Proposition 3.1 (Same (λ, μ, ρ) , different geometry). *There exist resolution geometries $\mathcal{G}_1 = (\Sigma_R, C_1, \Sigma_N)$ and $\mathcal{G}_2 = (\Sigma_R, C_2, \Sigma_N)$ with identical λ , μ , and ρ but different value of the gauge invariant $\text{tr}(\Sigma_R C \Sigma_N C^\top)$; hence $\mathcal{G}_1$ and $\mathcal{G}_2$ are not gauge-equivalent.*

Proof. Take $\Sigma_R = \text{diag}(3, 1)$, $\Sigma_N = \text{diag}(2.5, 0.7)$ (distinct spectra, top stratum). Let $K_1 = U_1 D V_1^\top$ and $K_2 = U_2 D V_2^\top$ be two whitened couplings with the *same* singular values $D = \text{diag}(0.8, 0.4)$ but different singular frames, and set $C_i = \Sigma_R^{1/2} K_i \Sigma_N^{1/2}$. By construction $\rho(\mathcal{G}_1) = \rho(\mathcal{G}_2) = (0.8, 0.4)$ and the spectra agree. The quantity $\text{tr}(\Sigma_R^a C \Sigma_N^b C^\top)$ is gauge invariant, since under (U, V) the conjugations by U cancel in the trace (Lemma 4.2). Numerically $\text{tr}(\Sigma_R C_1 \Sigma_N C_1^\top) = 4.803$ while $\text{tr}(\Sigma_R C_2 \Sigma_N C_2^\top) = 23.272$. A gauge invariant that differs on two configurations certifies they lie in different orbits. $\square$

Remark 3.2. The counterexample is not exotic: it is the generic situation. Fixing the spectra and the canonical correlations still leaves the coupling free to point in different directions relative to the eigen-axes of Σ_R and Σ_N , and that freedom is Δ -dimensional (Theorem 2.1).

4 The missing invariant: frame alignment

We now name what the coordinates missed.

Definition 4.1 (Frame-alignment invariants). The *frame-alignment invariants* of (Σ_R, C, Σ_N) are

$$m_{ab} = \text{tr}(\Sigma_R^a C \Sigma_N^b C^\top), \quad a = 0, \dots, p-1, \quad b = 0, \dots, q-1.$$

Lemma 4.2 (Gauge invariance). *Each m_{ab} is invariant under $O(p) \times O(q)$.*

Proof. Under (U, V) : $\Sigma_R^a \mapsto U \Sigma_R^a U^\top$, and $C \Sigma_N^b C^\top \mapsto U C V^\top (V \Sigma_N^b V^\top) V C^\top U^\top = U (C \Sigma_N^b C^\top) U^\top$. Hence $\Sigma_R^a C \Sigma_N^b C^\top \mapsto U (\Sigma_R^a C \Sigma_N^b C^\top) U^\top$, and the trace is unchanged. $\square$

Remark 4.3 (What the invariant means: magnitude vs. orientation). The canonical correlations $\rho_i = \cos \theta_i$ record the *magnitude* of the coupling—the principal angles between the resolved subspace and the informative directions, i.e. how strongly resolved and null uncertainties are tied. The frame-alignment invariants record the coupling’s *orientation*: whether that coupling is channelled through the high-variance or the low-variance axes of the resolved and null uncertainty ellipsoids. Two systems can couple equally strongly (same ρ) while one feeds the coupling into a stiff, well-measured resolved direction and the other into a soft, poorly-measured one—observably different for what the blind sector reveals about the resolved sector. The missing invariant is therefore not a bookkeeping artifact; it is real, observationally meaningful structure.

5 The repair: re-saturation

Adding the frame-alignment invariants restores completeness, and exactly—neither too few nor too many.

Theorem 5.1 (Re-saturation). *The invariant set*

$$\mathcal{I} = \{ \text{tr} \Sigma_R^k \ (k = 1, \dots, p), \quad \text{tr} \Sigma_N^k \ (k = 1, \dots, q), \quad m_{ab} \ (a < p, \ b < q) \}$$

has $p + q + pq$ members. At a generic point its differential has rank $p + q + pq$, equal to the quotient dimension. Consequently $\mathcal{I}$ locally separates gauge orbits: the fibre is saturated by $\mathcal{I}$, and there are no further hidden continuous invariants.

Proof. The count is immediate: $p + q + pq$. Each element is gauge invariant (power sums by the spectral theorem; m_{ab} by Lemma 4.2), so the differential of the map $\mathcal{I}$ annihilates the gauge-orbit directions and therefore has rank at most the quotient dimension $p + q + pq$ (Theorem 2.1). Computing the differential at a generic point gives rank exactly $p + q + pq$ (Verification 5.2); a map into $\mathbb{R}^{p+q+pq}$ whose differential has rank equal to the quotient dimension is a local submersion onto the quotient, hence locally injective on orbit classes. So $\mathcal{I}$ is a complete local invariant, and since its rank already equals the quotient dimension no additional invariant can raise it—the set is complete and not redundant. $\square$

Verification 5.2 (Re-saturation, measured). The Jacobian rank of $\mathcal{I}$ equals the quotient dimension at every tested size:

(p, q)	$ \mathcal{I} = p + q + pq$	Jacobian rank	quotient dim	saturated?
(2, 2)	8	8	8	yes
(3, 3)	15	15	15	yes
(2, 3)	11	11	11	yes
(4, 4)	24	24	24	yes

The equality (rank = quotient dim, not less and not more) is the two-sided statement the test was designed to produce: *under*-saturation by (λ, μ, ρ) is repaired, and no *over*-saturation (spurious extra invariant) is introduced.

Corollary 5.3 (Corrected fibre coordinates). *A resolution geometry, up to gauge, is determined on the top stratum by (λ, μ) together with the frame-alignment invariants $\{m_{ab}\}$, equivalently by the coupling expressed in the block eigenframes modulo signed permutations. The canonical correlations ρ are the sub-collection of these invariants insensitive to frame orientation.*

6 Reconciliation with the corpus

The gap does not contradict the earlier statement that canonical correlations classify the coupling on a stratum; it locates precisely where that statement holds.

Proposition 6.1 (ρ is complete on the isotropic stratum). *On the isotropic stratum $\Sigma_R = aI_p$, $\Sigma_N = bI_q$, the gauge quotient has dimension $2 + \min(p, q)$, and (a, b, ρ) is a complete invariant; the frame-alignment invariants are constant there.*

Proof. With Σ_R, Σ_N scalar the gauge fixes them and acts on C alone by $C \mapsto UCV^\top$, whose orbits are classified by the singular values of C (i.e. ρ up to the scalars). The free parameters are a, b and C (pq), the C -orbit has the singular-value normal form, so the quotient dimension is $1 + 1 + \min(p, q)$, matched by (a, b, ρ) . Since there is no distinguished eigenframe ($\Sigma_R^a \propto I$), $m_{ab} \propto \text{tr}(C\Sigma_N^b C^\top)$ reduces to functions of the singular values—no orientation content remains. Verified: quotient dimension $2 + \min(p, q)$ for $(p, q) = (2, 2), (3, 3), (4, 4)$. $\square$

Remark 6.2 (Where completeness lives). The completeness of ρ is a property of the *maximally degenerate* stratum, where the eigenframes collapse and only the coupling’s magnitude survives. On the top stratum, where the eigenframes are sharp, orientation becomes visible and ρ alone is deficient by Δ . The stratified picture is thus internally consistent: the invariant content changes from stratum to stratum, and the earlier “complete on a stratum” claim was the isotropic-stratum truth.

7 What this says about the final geometry

The saturation test returns a clean verdict on the object drawn previously.

- (1) **The fibre is saturated—by the corrected coordinates.** The complete continuous invariant of a fibre is $(\lambda, \mu, \{m_{ab}\})$, dimension $p + q + pq$ (Theorem 5.1), not (λ, μ, ρ) .
- (2) **No instability signals a further missing piece.** Because the corrected Jacobian rank equals the quotient dimension exactly, there is no residual direction along which an unspecified structure behaves like a core invariant: the local invariant ring is closed at $p + q + pq$.
- (3) **Base structure remains separate.** Saturation here is fibre-wise and local. The global monodromy $\rho : \pi_1 \rightarrow \text{Aut}(\mathcal{G})$ and its character variety are base-direction invariants, not part of the fibre count; full description of the bundle is the corrected fibre invariants together with the monodromy.

Correction 7.1 (Amends the visual-space paper). The coordinate statement “a resolution geometry modulo gauge is determined by (λ, μ, ρ) ” (Proposition “Fibre coordinates” in the stratified-space paper) holds only on the isotropic stratum and for $p = q = 1$. On the top stratum it under-counts by $\Delta = pq - \min(p, q)$; the complete list adds the frame-alignment invariants $\{m_{ab}\}$. This correction is recorded as the test working: the geometry made the missing invariant findable, and the invariant is now supplied.

8 Non-claims

Non-claim 8.1. Saturation here is of *continuous, local* invariants on a stratum. Discrete invariants (the residual signed-permutation gauge) and global invariants (monodromy, character variety) are separate and unaffected.

Non-claim 8.2. The frame-alignment invariants are inherited invariant theory (traces of products of a pair of forms and a matrix under $O(p) \times O(q)$), not a new primitive. The contribution is locating them as core fibre structure, consistent with the elimination verdict.

Non-claim 8.3. No physical claim is made. “Magnitude vs. orientation” of the coupling is an estimation-geometry statement about resolved–null inference, not a physical field.

9 Conclusion

The test the geometry made possible has paid for itself. Asked whether (λ, μ, ρ) saturates the fibre, the answer is no: the gauge quotient is $pq - \min(p, q)$ dimensions larger, and two geometries with identical spectra and canonical correlations can differ in $\text{tr}(\Sigma_R C \Sigma_N C^\top)$. The missing structure is the alignment of the coupling with the uncertainty axes—its orientation, as against the magnitude that ρ records. Adding the frame-alignment invariants m_{ab} re-saturates the fibre exactly, with the Jacobian rank meeting the quotient dimension and going no further, so nothing else is missing and nothing spurious is added. The completeness of ρ survives where it always lived, on the isotropic stratum, and the stratified object is consistent throughout. A geometry earns the name partly by supporting this kind of audit; resolution geometry did, failed the naive version, and passes the corrected one.

A Numerical verifications

All computations use `numpy`. Gauge orbit dimensions are the rank of the infinitesimal action $(A, B) \mapsto ([A, \Sigma_R], AC - CB, [B, \Sigma_N])$ on $\mathfrak{so}(p) \oplus \mathfrak{so}(q)$; invariant-map Jacobians are finite-difference (step 10^{-6} , rank tolerance 10^{-4}).

- **Quotient dimension (§2).** Measured quotient dimension = $p + q + pq$ and gap $\Delta = pq - \min(p, q)$ for $(p, q) = (1, 1), (2, 2), (3, 3), (2, 3), (4, 4)$: $\Delta = 0, 2, 6, 4, 12$.
- **Counterexample (§3).** $\Sigma_R = \text{diag}(3, 1)$, $\Sigma_N = \text{diag}(2.5, 0.7)$, $\rho = (0.8, 0.4)$ for both; $\text{tr}(\Sigma_R C_1 \Sigma_N C_1^\top) = 4.803$, $\text{tr}(\Sigma_R C_2 \Sigma_N C_2^\top) = 23.272$.
- **Re-saturation (§5).** Jacobian rank of $\mathcal{I}$ equals $p+q+pq$ for $(p, q) = (2, 2), (3, 3), (2, 3), (4, 4)$: ranks 8, 15, 11, 24.
- **Isotropic stratum (§6).** Quotient dimension $2 + \min(p, q)$ for $(2, 2), (3, 3), (4, 4)$: 4, 5, 6, matched by (a, b, ρ) .

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